TinySAM: Exploratory Data Analysis

Oxford-IIIT Pets | Raw dataset audit | Python 3.14

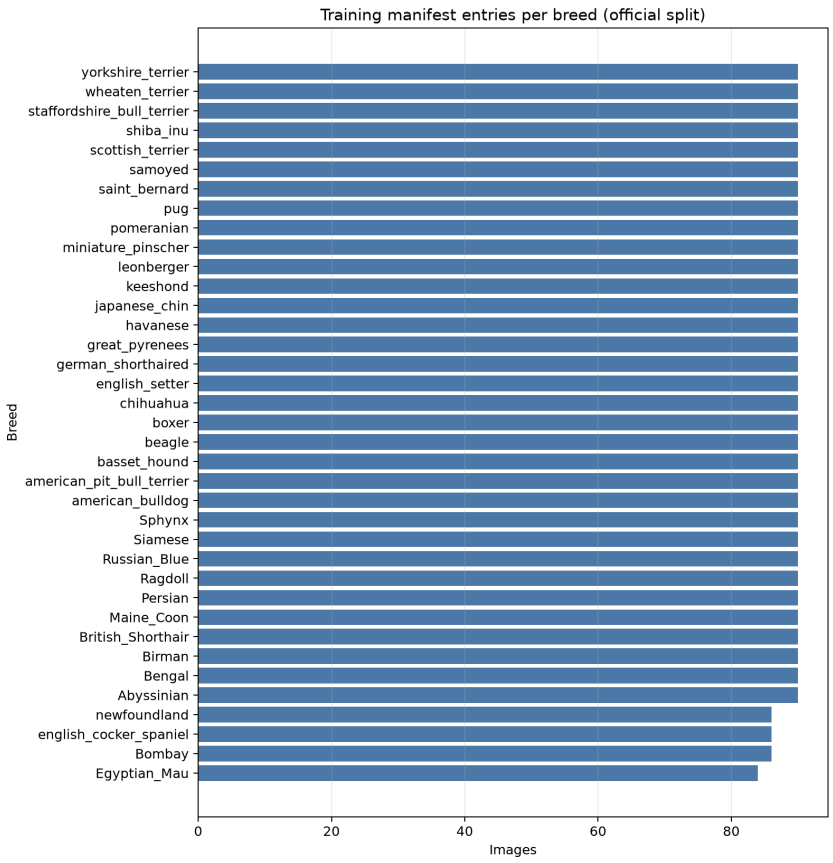
1. Dataset overview and split integrity

TinySAM is evaluated as a promptable **binary foreground-segmentation** system on Oxford-IIIT Pets. The supplied annotation README defines label 1 as foreground, label 2 as background, and label 3 as not classified. The project policy converts the trimap conceptually to `trimap != 2`, thereby including the not-classified label in its foreground target. This report does not apply that conversion to a cache; it uses it only to describe the label distribution and make that policy explicit.

Split	Expected pairs
test	3,669
train	3,312
val	368

The official `trainval.txt` list is divided deterministically into the project train/validation split (every tenth entry becomes validation), while `test.txt` remains held out. The raw manifests contain **7,349 samples**, **37 breeds**, and two species (cat: 2,371, dog: 4,978). In the training split, the smallest breed has 84 images and the largest has 90; the max/min imbalance ratio is 1.07.

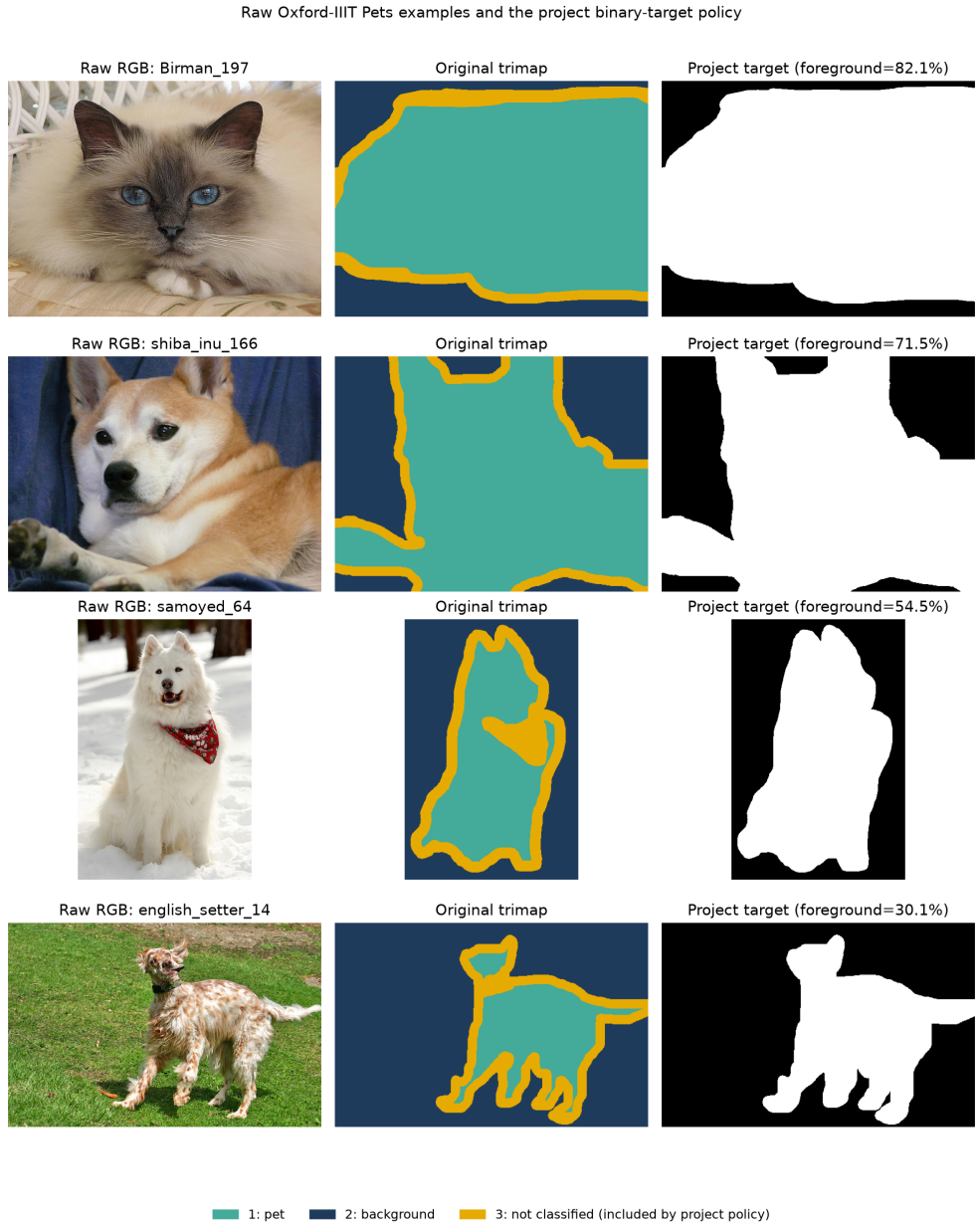
Integrity checks found 0 missing image files, 0 missing trimaps, 0 unreadable pairs, 3 non-RGB images, 0 image/mask dimension mismatches, and 0 trimaps with labels outside {1, 2, 3}. The 0 readable pairs with invalid labels or mismatched dimensions are retained in the audit but excluded from quantitative image/mask summaries. There are 0 image IDs shared by the official train/validation and test lists.



Training images per breed

2. Raw visual inspection and label balance

The examples below show the unmodified RGB image, the original three-value trimap, and the binary target used by the project. Because label 3 is not classified rather than a confirmed foreground boundary, including it in the project target is a documented modelling choice that must remain consistent in preprocessing and evaluation.

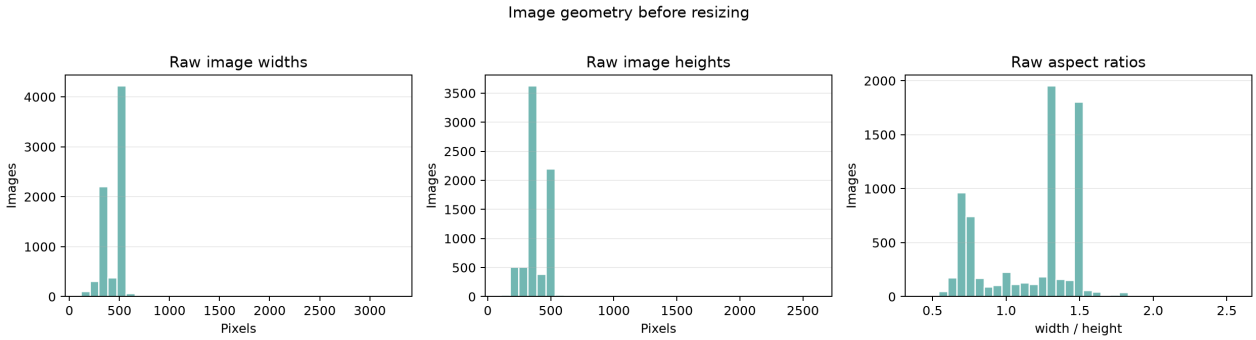


Raw examples, trimaps, and derived targets

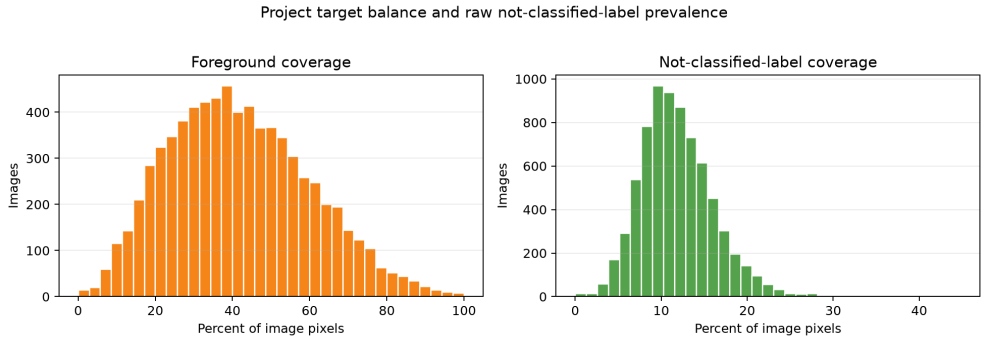
Across valid, dimension-matched pairs, project-target foreground coverage has median 40.38% (mean 41.82%); strict label-1 foreground coverage has median 27.41%, and the raw not-classified label has median coverage 11.36%. 15 masks have project-target foreground coverage below 1% or above 99%. Pixel accuracy alone would therefore be misleading when background dominates an image.

3. Image geometry and data challenges

Raw image widths range from 114 to 3264 pixels and heights from 103 to 2606 pixels. Aspect ratios range from 0.42 to 2.56. The varying source geometry motivates aspect-distorting square resizing as an explicit, documented preprocessing trade-off rather than an unnoticed assumption.



Raw image geometry



Project foreground and not-classified-label coverage

The data audit also identifies 14 empty masks, 0 all-foreground masks, and 14 masks that cannot support both positive and negative point prompts. These cases must be handled transparently in the later prompt-sampling pipeline; they are not removed in this EDA.

4. Evaluation metrics

The primary test metric will be **per-image foreground IoU**, averaged over the held-out test images (mIoU). It directly measures mask overlap and avoids the inflated impression that a background-heavy pixel-accuracy score can give. **Dice/F1** will be reported alongside IoU because it is especially informative when foreground objects are small. To make class imbalance visible, report macro breed/species summaries in addition to the overall mean. Finally, because TinySAM is promptable, report point-prompt mIoU and box-prompt mIoU separately, using a fixed random seed and the same held-out split.